or $y\left(1+x^3\right)^2 = x+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^6}{6}+C$

and $y = \frac{1}{(1+x^3)^2}\left[x+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^6}{6}+C\right]$ is the required solution.

Example 10.26

Solve $ye^y dx = \left(y^3+2xe^y\right)dy$.

Solution

The given equation can be written as $\frac{dx}{dy}-\frac{2}{y}x = y^2e^{-y}$.

This is a linear differential equation. Here $P = -\frac{2}{y}$; $Q = y^2e^{-y}$.

$$\int P\,dy = \int -\frac{2}{y}dy = -2\log|y| = \log|y|^{-2} = \log\left(\frac{1}{y^2}\right),$$

$$\text{Thus, I.F.} = e^{\int P dy} = e^{\log\left(\frac{1}{y^2}\right)} = \frac{1}{y^2}.$$

$$\text{Hence the solution is } xe^{\int P dy} = \int Qe^{\int P dy}dy + C$$

$$\text{That is, } x\left(\frac{1}{y^2}\right) = \int y^2e^{-y}\left(\frac{1}{y^2}\right)dy + C = \int e^{-y}dy + C = -e^{-y} + C$$

$$\text{or } x = -y^2e^{-y} + Cy^2 \text{ is the required solution.}$$

EXERCISE 10.7

Solve the following Linear differential equations:

1. $\cos x\frac{dy}{dx} + y\sin x = 1$
2. $\left(1-x^2\right)\frac{dy}{dx} - xy = 1$
3. $\frac{dy}{dx} + \frac{y}{x} = \sin x$
4. $\left(x^2+1\right)\frac{dy}{dx} + 2xy = \sqrt{x^2+4}$
5. $\left(2x-10y^3\right)dy + ydx = 0$
6. $x\sin x\frac{dy}{dx} + \left(x\cos x + \sin x\right)y = \sin x$
7. $\left(y - e^{\sin^{-1}x}\right)\frac{dx}{dy} + \sqrt{1-x^2} = 0$
8. $\frac{dy}{dx} + \frac{y}{(1-x)\sqrt{x}} = 1-\sqrt{x}$
9. $\left(1+x+xy^2\right)\frac{dy}{dx} + \left(y+y^3\right) = 0$
10. $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$
11. $(x+a)\frac{dy}{dx} - 2y = (x+a)^4$
12. $\frac{dy}{dx} = \frac{\sin^2 x}{1+x^3} - \frac{3x^2}{1+x^3}y$
13. $x\frac{dy}{dx} + y = x\log x$
14. $x\frac{dy}{dx} + 2y - x^2\log x = 0$
15. $\frac{dy}{dx} + \frac{3y}{x} = \frac{1}{x^2}$, given that $y = 2$ when $x = 1$

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